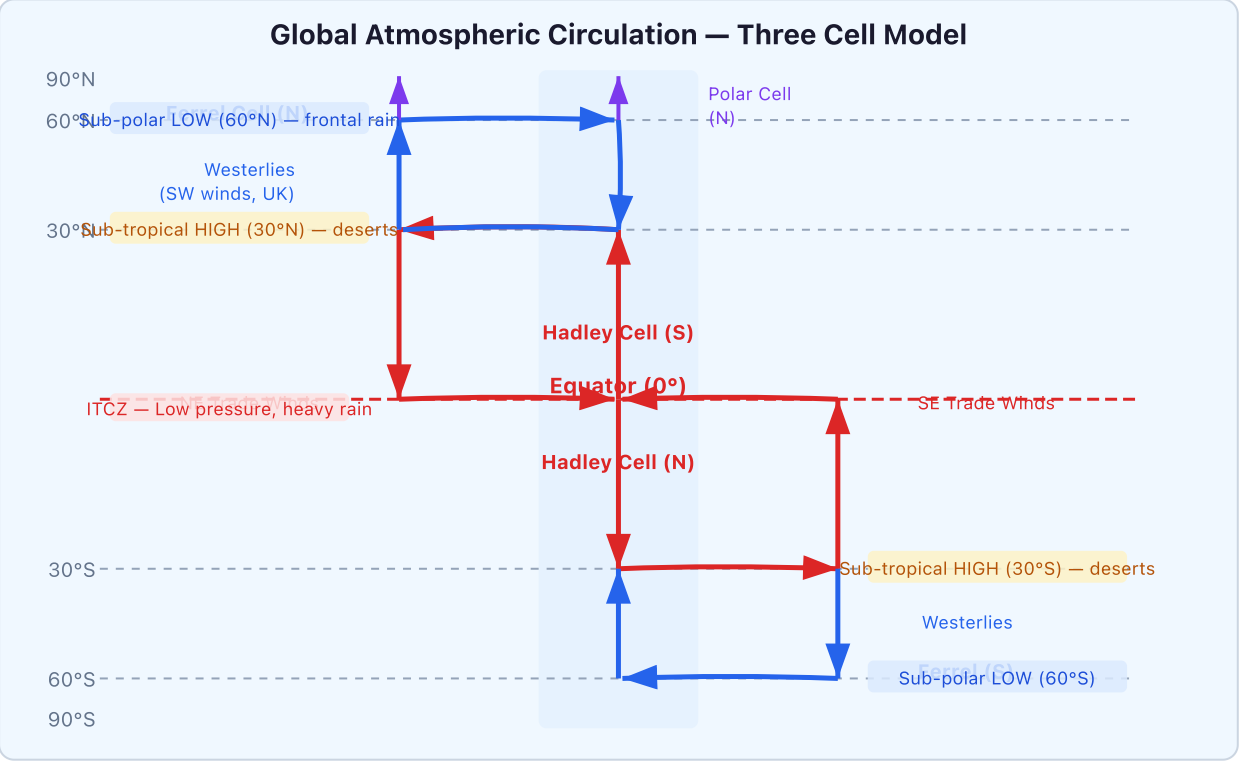
OCR GCSE Geography B (J384) — Paper 1: Our Natural World · Topic 1: Global Hazards

1.1 Global Hazards: Extreme Weather

1. Global Atmospheric Circulation

Global atmospheric circulation: The large-scale movement of air around the Earth driven by unequal heating. The equator receives more solar radiation than the poles, creating temperature and pressure differences. This drives a system of convection cells (Hadley, Ferrel, and Polar cells) that redistribute heat energy around the globe and create the major wind belts and climate zones.



Pressure Belt / Zone	Latitude	Conditions	Climate result
ITCZ (Inter-Tropical Convergence Zone)	0° (equator)	Rising air, intense heating, low pressure	Equatorial climate: hot, very wet all year; tropical rainforest

Sub-tropical high pressure	~30°N/S	Descending air, high pressure, dry	Hot desert climates (Sahara, Arabian, Australian deserts)
Sub-polar low pressure	~60°N/S	Cold polar air meets warm tropical air; fronts; rising air	Temperate climate with frontal rainfall (UK, NW Europe)
Polar high pressure	90°N/S	Descending cold air, high pressure, very dry	Polar/tundra climate: very cold, low precipitation (mainly snow)

Surface winds

- **Trade winds** blow from the sub-tropical high (30°) back towards the low pressure at the Equator. The Coriolis effect bends them, so they blow from the north-east in the Northern Hemisphere and south-east in the Southern Hemisphere.
- **Westerlies** (south-westerlies over the UK) blow from the sub-tropical high towards the sub-polar low at 60°, bringing mild, wet Atlantic air and depressions to the UK.
- Where trade winds meet at the ITCZ, warm moist air rises, giving heavy convectional rain — the same rising air that helps tropical storms form over warm oceans.

2. El Niño and La Niña

The **El Niño Southern Oscillation (ENSO)** is a natural cycle of changes in ocean temperatures and winds across the Pacific Ocean, occurring every 2–7 years.

	Normal conditions	El Niño	La Niña
Trade winds	Blow east to west across the Pacific	Weaken or reverse	Stronger than normal
Warm water	Piles up in the western Pacific (near Australia/Indonesia)	Spreads east towards South America	Pushed even further west
Australia / Indonesia	Rising air and rain	Sinking air — **drought** and wildfires	Heavy rain and **flooding**

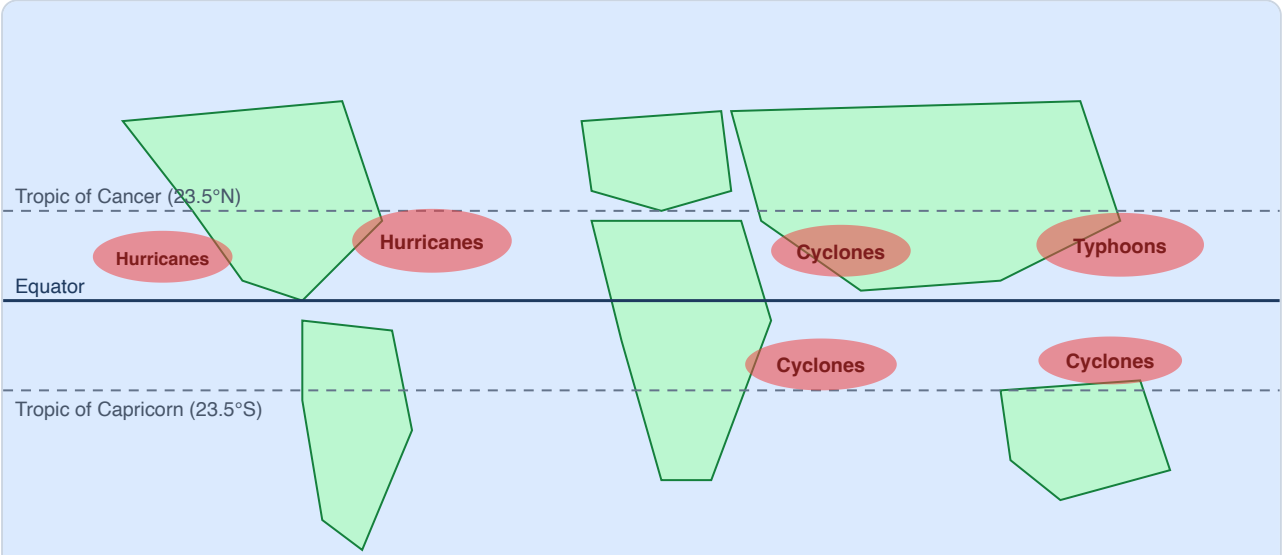
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Peru / western South America	Cold upwelling water, dry	Heavy rain and **flooding**; fish stocks collapse	Drier, cooler than normal

Exam Tip: El Niño and La Niña show how changes in atmospheric and ocean circulation can cause extreme weather thousands of kilometres apart — drought in one place and floods in another at the same time.

3. Global Distribution of Tropical Storms

Tropical storm — an intense low-pressure weather system with sustained wind speeds above 119 km/h (to count as a hurricane, typhoon or cyclone), heavy rainfall and a calm central eye. They are called **hurricanes** in the North Atlantic and NE Pacific, **typhoons** in the NW Pacific, and **cyclones** in the Indian Ocean and South Pacific.



Tropical storms form between about 5° and 30° north and south of the Equator (simplified map, not to scale)

- Tropical storms form over **warm tropical oceans**, mostly between **5° and 30° north and south** of the Equator.
- They do not form **within about 5° of the Equator** because the Coriolis effect is too weak there to make the air spin.

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- They move **westwards** with the trade winds, then often curve polewards. They **weaken over land** (no supply of warm, moist air) and over cooler water.
- Most form in **late summer and autumn** (June–November in the Northern Hemisphere), when sea temperatures are highest.

4. Causes and Formation of Tropical Storms

Conditions needed

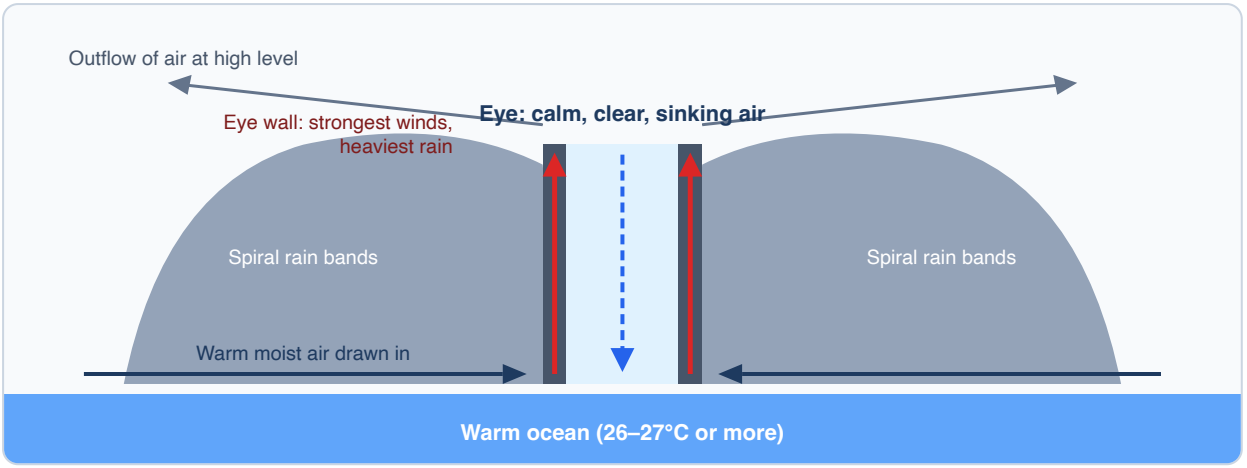
- Ocean surface temperature of **at least 26–27°C**, to a depth of about 50–60 m.
- Ocean **at least 5° from the Equator**, so the Coriolis effect can make the storm spin.
- **Low wind shear** — winds at different heights blowing at similar speeds, so the storm's structure is not torn apart.
- An area of **low pressure** with rising, unstable, moist air.

Sequence of formation

1. Warm ocean water heats the air above it. The **warm, moist air rises rapidly**, creating an area of very low pressure at the surface.
2. As the air rises it **cools and condenses**, forming towering cumulonimbus clouds. Condensation releases **latent heat**, which warms the air further and makes it rise even faster — powering the storm.
3. Air rushes in at the surface to replace the rising air, creating strong winds. The **Coriolis effect** makes the inflowing air spiral (anticlockwise in the Northern Hemisphere).
4. Some cool, dry air sinks down the centre of the storm, creating the calm, cloud-free **eye**.
5. The storm is carried across the ocean by prevailing winds, growing as long as it has a supply of warm, moist air. It weakens when it reaches **land or cooler water** because the energy supply is cut off.

Structure of a tropical storm

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Feature	Description
Eye	Centre of the storm, 10–50 km across. Calm, clear skies, lowest pressure, caused by sinking air.
Eye wall	Ring of towering cumulonimbus cloud around the eye — the most intense winds (often over 200 km/h) and heaviest rainfall.
Rain bands	Spiralling bands of cloud and rain extending hundreds of kilometres from the centre.
Size	Typically 200–700 km across; can last for 1–3 weeks.

5. Physical Hazards of Tropical Cyclones

Hazard	Effect
High winds	Sustained winds over 119 km/h (over 250 km/h in the strongest storms) destroy buildings, uproot trees and turn debris into missiles.
Intense rainfall	Hundreds of millimetres of rain in a day or two causes river and flash flooding.
Storm surges	Low pressure and onshore winds raise sea level by several metres — usually the most deadly hazard.

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Coastal flooding	Surges combined with high tides and waves flood low-lying coasts, contaminating land and water with salt.
Landslides	Saturated slopes collapse, burying homes and blocking roads.

Cyclone strength is measured on the **Saffir–Simpson scale** from category 1 (119–153 km/h) to category 5 (252 km/h or more).

6. How Climate Change Might Affect Tropical Storms

Aspect	Likely change	Reason
Distribution	Storms may form further from the tropics (further north and south)	A larger area of ocean will reach 26–27°C
Frequency	Total number may stay the same or fall slightly	Scientists are uncertain — changes in wind shear may suppress some storms
Intensity	A higher proportion of storms reaching the strongest categories (4 and 5), with heavier rainfall	Warmer oceans supply more energy; warmer air holds more moisture
Storm surges	More damaging coastal flooding	Sea level rise means surges start from a higher base

Exam Tip: Be careful with certainty. The evidence for *more intense* storms is much stronger than the evidence for *more frequent* storms — say "likely" or "scientists predict" rather than "will".

7. Case Study: Typhoon Haiyan, the Philippines, November 2013

Typhoon Haiyan (known locally as Yolanda) — 8 November 2013

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Background: The Philippines is a lower-middle-income country made up of over 7,000 islands in the western Pacific — one of the most typhoon-exposed countries in the world. Haiyan was one of the strongest tropical storms ever to make landfall, with sustained winds of about **315 km/h**. It hit the islands of Samar and Leyte, including the city of **Tacloban**.

Primary effects

- A **storm surge of up to 5–6 m** swept through Tacloban, flooding the city and its airport terminal.
- Over **6,300 people killed**; around 1,000 more missing; nearly 30,000 injured.
- About **1.1 million homes damaged or destroyed** — around 90% of Tacloban was destroyed.
- Around 4 million people displaced; power lines, roads and bridges destroyed.
- About 400 mm of rain caused widespread flooding.

Secondary effects

- About 14 million people affected; around 6 million workers lost their source of income.
- Flooding caused **landslides** and blocked roads, cutting off aid to remote areas.
- Shortages of food, water and shelter led to **outbreaks of disease** and some looting.
- Rice harvests and fishing fleets destroyed, causing food shortages; coconut plantations (a major source of income) flattened.
- An oil barge ran aground causing an oil spill, damaging mangroves.

Immediate responses

- The government evacuated around **800,000 people** before landfall, using warnings from the national weather agency (PAGASA).
- International aid and teams arrived — the UK sent military aircraft and HMS Daring; the USA sent the aircraft carrier USS George Washington to supply water and food.
- Field hospitals set up; emergency shelters, food, water and hygiene kits distributed by NGOs such as the Red Cross and Oxfam.
- A curfew was imposed two days after the storm to reduce looting.

Long-term responses

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- A **"Build Back Better"** reconstruction plan — rebuilding roads, bridges and airports to a higher standard.
- "Cash for work" programmes paid local people to clear debris and rebuild, restoring incomes.
- Rice farmers and fishermen given help to restart — seeds, tools and replacement boats.
- A **"no-build zone"** proposed along vulnerable coasts, with communities relocated inland.
- More **cyclone shelters** built; storm-surge warnings improved so people understood the risk (many did not know what a "storm surge" meant in 2013).
- Mangroves replanted along the coast to act as a natural barrier against future surges.

8. Reducing the Effects of Tropical Storms

Strategy	How it reduces the effects	Example
Monitoring	Satellites, aircraft and weather buoys track the storm's position, wind speed and pressure in real time	US National Hurricane Center tracks every Atlantic hurricane
Prediction	Computer models forecast the storm's path and strength several days ahead, allowing warnings and evacuation	PAGASA warnings allowed 800,000 to evacuate before Haiyan
Protection	Storm-surge barriers, sea walls, raised buildings on stilts, reinforced roofs, cyclone shelters	Bangladesh has built thousands of raised cyclone shelters, cutting death tolls dramatically since 1991
Planning	Evacuation routes, emergency supplies, education and drills; land-use zoning to keep homes away from the coast	Florida's signed hurricane evacuation routes; Philippines' no-build zones

Exam Tip: Planning and prediction save the most lives at the lowest cost — Bangladesh's 1970 Bhola cyclone killed around 300,000–500,000 people, whereas Cyclone Sidr (2007), of similar

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strength, killed about 3,400 thanks to early warnings and shelters. Use this contrast to evaluate strategies.

9. Drought as a Weather Hazard

Drought — a long period of abnormally low rainfall that leads to a shortage of water. Droughts develop slowly and can last months or years.

- **Causes:** persistent high pressure (sinking air prevents clouds forming); changes to atmospheric circulation such as El Niño; shifts in the ITCZ; human activity such as over-abstraction and deforestation can make the effects worse.
- **Distribution:** most common in sub-tropical areas around 30° north and south (south-west USA, Australia, the Sahel, southern Africa), but can affect anywhere, including the UK.

10. Case Study: The Millennium Drought, Australia (1997–2009)

The Millennium Drought — south-east Australia

Causes: repeated El Niño events and persistent high pressure over southern Australia reduced rainfall for over a decade; rising temperatures increased evaporation.

Impacts

- **Environmental:** river flows in the **Murray–Darling Basin** fell to record lows; wetlands dried out; fish and bird populations declined; dust storms.
- **Economic:** farm output fell sharply — rice and cotton production collapsed in some years; the drought cost the economy billions of dollars and thousands of farming jobs.
- **Social:** strict water restrictions in cities such as Melbourne and Brisbane; stress and financial hardship for farming families; some rural towns declined.
- Dry conditions contributed to the **Black Saturday bushfires** (Victoria, February 2009), which killed 173 people.

Responses

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- Water restrictions — hosepipe bans and limits on garden watering; targets such as Melbourne's 155 litres per person per day.
- Large **desalination plants** built in Perth, Sydney, Melbourne, Adelaide and on the Gold Coast.
- The **Murray–Darling Basin Plan** (2012) set limits on how much water can be taken for irrigation, returning water to rivers.
- Water recycling, rainwater tanks and more efficient irrigation on farms.

Exam Tip: Compare a drought in an Advanced Country (Australia) with a tropical storm in an LIDC/EDC (Haiyan) — wealthier countries can afford expensive responses such as desalination, but drought still causes serious economic and environmental damage.

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